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Autonomous University of Ciudad Juárez

Institute of Engineering and Technology

Department of Electrical and Computer Engineering

Task 16 Line-following cart

Sensors and Instrumentation

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Index

1. Objective 2

2. Introduction 2

3. Definitions and conceptual framework 3
   3.1. Photodiode. 3
   3.2. Phototransistor. 3
   3.3. Direct Current (DC) Motors 4
   3.4. PWM Signals 4
   3.5. Bridge H (L298N) 5

4. Development 6

5. Code 7

6. Results 9

7. Conclusion 10

1. Objective

Develop a functional prototype of a line-following cart using sensors infrared and DC motors controlled by PWM, using a ESP32 microcontroller. The system must be able to detect a trajectory in a track and follow it autonomously, demonstrating the principle of feedback between sensors and actuators.

2. Introduction

A line follower is a type of mobile robot that uses optical sensors to detect and follow a defined path, usually a line of contrasting color (black on (white background). These systems are one of the most common applications in robotics educational due to its ability to demonstrate fundamental concepts of electronics, control and programming in a practical way.

In this task, a prototype of a line-following cart was developed using a microcontroller. ESP32, infrared sensors consisting of a photodiode-phototransistor pair, and motors of Direct current controlled by pulse-width modulation (PWM). The system It was designed to follow a track, reacting to changes in ground reflectivity. to adjust the vehicle's movement in real time.

3. Definitions and conceptual framework

3.1. Photodiode

The FD-IR333C is an infrared light emitting diode that typically operates at a voltage direct current of approximately 1.2V. Its function is to emit infrared radiation which will reflected by the surface under the sensor. Light surfaces reflect more light, while that dark cells absorb a large part of the radiation. In this project, the photodiode is connects in series with a 100 ohm current-limiting resistorΩ,forming part of sensor pair that detects the line on the track.

Figure 1: Transparent Photo Diode (FD-IR333C)

3.2. Phototransistor

The PT331C is a phototransistor that responds to infrared radiation emitted by the photodiode. When reflected infrared light strikes its base, the phototransistor allows the flow of current from the collector to the emitter. In this configuration, the collector is connected to a 10k pull-up resistorΩtowards 5V and also to the analog input of the ESP32, while the emitter is connected to ground. This arrangement allows for the detection of changes in the amount of reflected light as variations in output voltage, which the microcontroller interprets to determine the position of the cart with respect to the line.

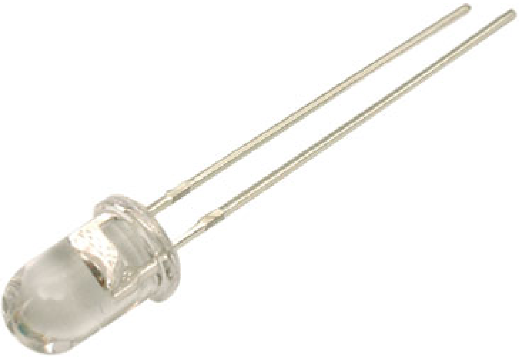

Figure 2: Phototransistor (PT331C)

3.3. Direct Current (DC) Motors

Direct current motors convert electrical energy into rotary motion.

They are widely used due to their ease of control and low cost. In this practice,

They used 6V motors, which allow for acceptable speeds for a

laboratory or classroom environment.

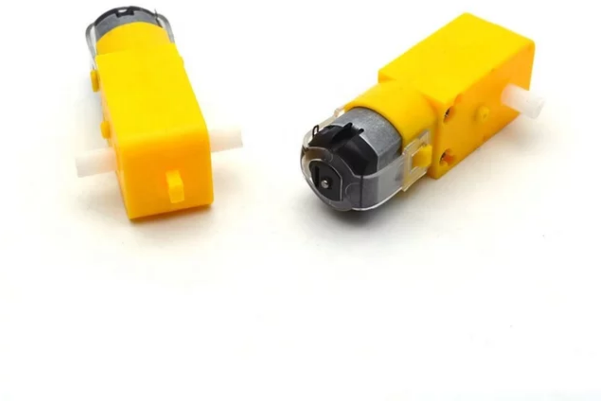

Figure 3: 6V DC Motor

3.4. PWM Signals

Pulse width modulation (PWM) is a technique that

It allows you to control the speed of a DC motor by adjusting the duty cycle.

cycle) of a digital signal. The higher the duty cycle, the greater the energy supplied to the engine, and therefore, its speed.

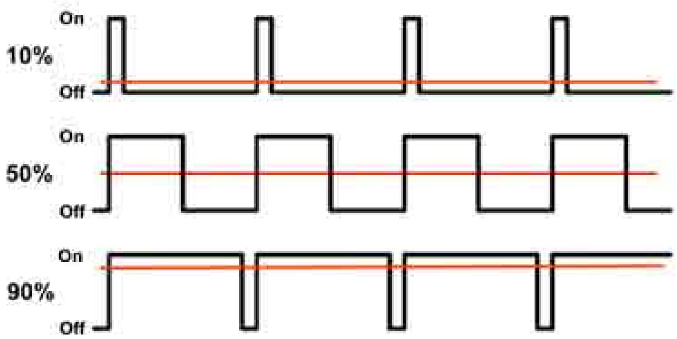

Figure 4: PWM Graph

3.5. Bridge H (L298N)

The H-bridge is a circuit that allows changing the polarity of the power supply to the motors, which allows their direction of rotation to be reversed. The L298N is a dual-bridge motor that It allows you to control two motors independently, and can also control their speed using the PWM signal sent from a microcontroller such as the Arduino.

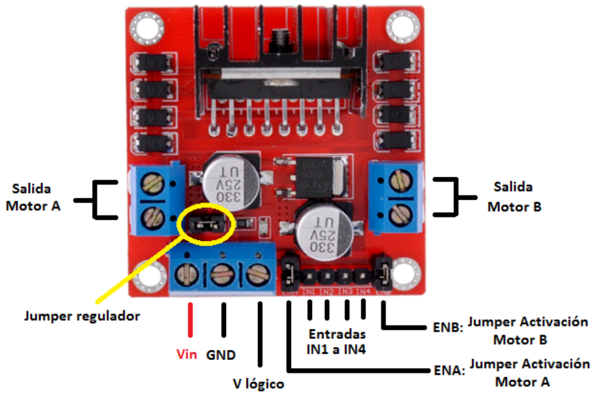
Salida Motor A
Salida Motor B
Jumper regulador
Vin
GND
V lógico
Entradas IN1 a IN4
ENB: Jumper Activación Motor B
ENA: Jumper Activación Motor A

Figure 5: H-bridge L298N

4. Development

For line detection, two sensor pairs were used: each consisting of a FD-IR333C photodiode and a PT331C phototransistor.

For the photodiodes, a 100 ohm limiting resistor is usedΩwith a diet of 5V and an approximate voltage drop of 1.2V, with both diodes in parallel which divides the current for each LED in half. Applying Ohm's Law:

$V_{led} = 5\,V - 1,2\,V = 3,8$

$Yo_{led} = \frac{V_{led}}{R} \times \frac{1}{2} = \frac{3,8\,V}{100} \times \frac{1}{2} \approx 20\,mA$

This configuration ensures stable operation and adequate current consumption to protect both the LEDs and the ESP32.

The phototransistors were connected with the collector to a 100k pull-up resistorΩ which goes to 5V, and also to an analog input of the ESP32 (GPIO34 and GPIO35). The emitter of each phototransistor is connected directly to ground. Thus, when the sensor It detects a light surface (which reflects more IR), the phototransistor conducts and the voltage across The input pin decreases. When it detects a black line (less reflection), conduction decreases and the input voltage rises, allowing the position of to be clearly distinguished the line.

The system has two direct current motors, one for each wheel. These motors They were connected directly to the GPIO16, GPIO17, GPIO18 and GPIO19 pins of the ESP32 through a driver that allows H-Bridge switching for each motor.

Each motor receives two signals: one PWM control signal for speed, and another logic signal for the direction of rotation. The left motor is controlled by pins 18 and 19, and the right one by pins 18 and 19. through pins 16 and 17.

The main algorithm consists of constantly reading the analog values of the two sensors and compare their values against a threshold (approximately 1900). If the sensor The left side detects a line and the right side does not, the cart turns left; if the opposite occurs, Turn right. If both sensors detect a line or bottom, the cart stops.

Motor speed is controlled by PWM, with a value of 255 (maximum speed).

5. Code

This was the code used, as explained above:

Listing 1: Line follower cart code

```
#include<Arduino.h>

// Sensors        analgesics Left
const   int       sensor          =  35;     //   GPIO34   (ADC1_CH6)
const   int    sensorDer          =  34;     //   GPIO35   (ADC1_CH7)

// Engine A (left wheel) const
        int    motorA_IN1         =  18;     //   GPIO18
const   int    motorA_IN2         =  19;     //   GPIO19

// Engine B (right wheel) const
        int    motorB_IN1         =  16;     //   GPIO16
const   int    motorB_IN2         =  17;     //   GPIO17

// ======== PARAMETERS ======== const
        int    THRESHOLD          =  1900;
const   int    SPEED              =  255;

voidsetup() {
   Serial.begin (115200);
   pinMode(sensorIzq ,        INPUT);
   pinMode(sensorDer ,        INPUT);
   pinMode(motorA_IN1 ,         OUTPUT);
   pinMode(motorA_IN2 ,         OUTPUT);
   pinMode(motorB_IN1 ,         OUTPUT);
   pinMode(motorB_IN2 ,         OUTPUT);
}

voidloop() {
   intleftval = analogRead(leftsensor);
   intvalDer = analogRead(sensorDer) -300;
```

```
    Serial.printf("Left Sensor = %4d         Sensor Der= %4d\n", valIzq , valDer);

if(leftval < THRESHOLD && rightval > THRESHOLD) {
        leftturn();
    }
    else if(leftval > THRESHOLD && rightval < THRESHOLD) {
        turn();
    }
    else{
        stall ();
    }
delay (1);
}

voidleftturn() {
    analogWrite(motorA_IN1 ,      0);          digitalWrite(motorA_IN2 ,       LOW);
    analogWrite(motorB_IN1 ,      SPEED);       digitalWrite(motorB_IN2 ,       LOW);
}

voidturn() {
    analogWrite(motorA_IN1 ,      SPEED);       digitalWrite(motorA_IN2,        LOW);
    analogWrite(motorB_IN1 ,      0);           digitalWrite(motorB_IN2,        LOW);
}

voidstall () {
    digitalWrite(motorA_IN1 ,       LOW);   digitalWrite(motorA_IN2 ,       LOW);
    digitalWrite(motorB_IN1 ,       LOW);   digitalWrite(motorB_IN2 ,       LOW);
}
}
```

6. Results

Upon completion of the implementation of the line-following cart, its functionality was verified. on a track with a black line on a white background. The cart was able to identify correctly the line and react according to the programmed logic. When the sensor When the left side detected the line, the cart turned left; when it was the right side, It turned to the right; and when both sensors were out of line, the cart It stopped. The response was quick and stable during testing.

The infrared sensors responded with clear and distinguishable readings: high values when there was no line (white background) and low values when detecting the black line. This It allowed a reliable detection threshold to be established in 1900, with minimal calibration. in addition to the right sensor (-300) to compensate for differences in sensitivity. Furthermore, The PWM control provided a constant and suitable speed for the test without causing no sudden movements or deviations off the track.

During testing, the cart completed multiple routes without losing its path or needing External intervention. The system proved efficient for basic monitoring tasks. in line, fulfilling the objectives established for the practice.

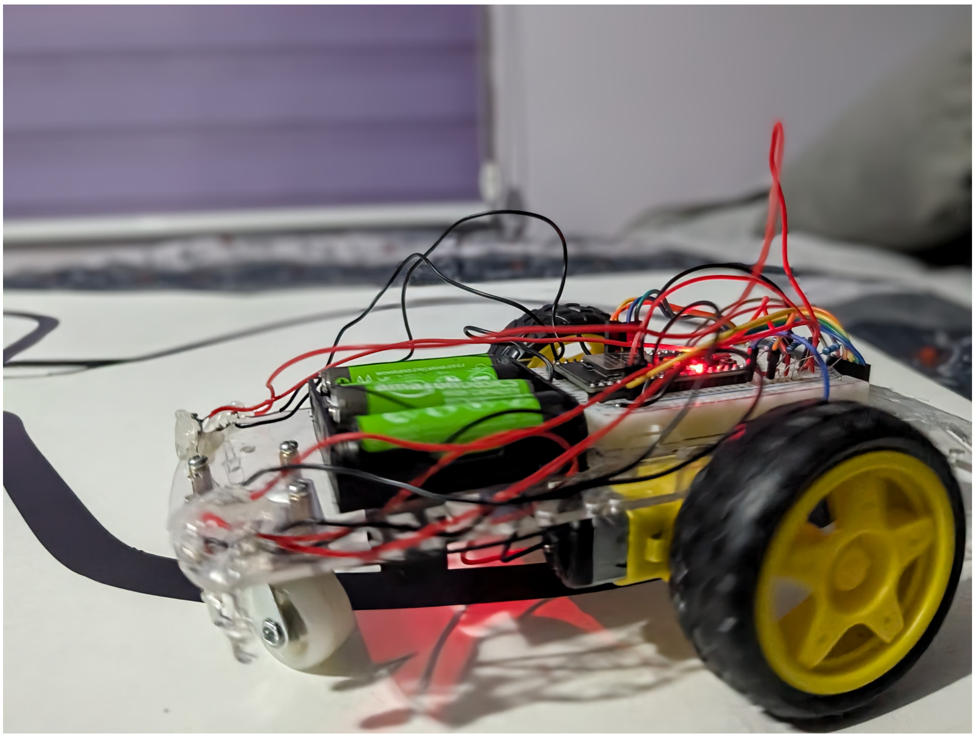

Figure 6: Cart in operation

7. Conclusion

The line-following cart project met the established objectives in a satisfactory. The combination of infrared sensors (FD-IR333C and PT331C), the microcontroller The ESP32 and direct current motors made it possible to create an autonomous navigation system capable of following a line with precision.

The use of pulse width modulation (PWM) facilitated efficient control of the Motor speed, which improved the cart's stability. Analog reading The infrared sensors allowed for the reliable detection of variations between background white and the black line, which made it possible for the cart to react correctly to the different situations that arose on the track.

The system showed consistent behavior in the tests, performing precise turns. and stopping when necessary. Although the system is functional, problems can arise. improvements, such as the implementation of adaptive cruise control or the incorporation of additional sensors to improve accuracy on tighter curves.

Overall, the project was successful in its implementation and represents a solid foundation. for future robotics and automation projects.